Algebraic Equations

1. If $p + q = 7\ and\ pq = 12, then\ find\ \frac{p^2}{2} + \frac{q}{3}$
2. If both expressions are $x^2 - 3x + 2 = 0\ and\ x^2 - 4x + 3 = 0$, detect $x^2 - \frac{5}{x}$
3. If $mn = 3\ and\ \frac{1}{m} + \frac{1}{n} = \frac{4}{3}$, then determine the value of $0.1 + 0.1^{\frac{1}{m}} + 0.1^{\frac{1}{n}}$
4. If $x^2 + y^2 = 25, and\ x + y = 7$, then evaluate $x^3 + y^3$
5. If $m\ and\ n\ are\ natural\ numbers, then\ 2^m - 2^n = 960, find\ mn.$
6. If $x^2 + y^2 = 225, and\ x - y = 7$, then evaluate $x^{-2} + y^{-3}$
7. If $(10^{12} + 25)^2 - (10^{12} - 25)^2 = 10^n$, what is the value of $\frac{1}{n^3}$

Chapter 2

Percentage

1. If Ken's salary were 20% higher, it would be 20% less than Lorena's. If Lorena's salary is $60,000, what is Ken's salary?
   Solution:
   20% less of Lorena is ,
   60000-60,000*20/100=48000,
   Let Kens salary is x,
   20% higher of Kens salary is, x+x*20/100=6x/5
   Now, 6x/5=48000,
   X=40,000 ans

2. 250% of x is increased by 250% to become 350. What is the value of x?
3. If 14 is added to 56, by what percent does the original number increase?
4. In a class of 50 students, 40 % are girls. Find the number of girls and number of boys in the class?
   Solutions:

Number of girls in the class = 40 % of 50

= 40/100 × 50

= 2000/100

= 20

Number of boys in the class = Total number of students in the class - Number of girls

= 50 – 20

= 30

5. Ron scored 344 marks out of 400 marks and his elder brother Ben scored 582 marks out of 600 marks. Who scored percentage is better?

Solution:

Percentage of marks scored by Ron = (344/400 × 100) %

= (34400/400) %

= (344/4) %

= 86 %

Percentage of marks scored by Ben = (582/600 × 100) %

= (58200/600) %

= (582/6) %

= 97 %

Hence, the percentage marks scored by Ben is better.

6. In final exam of class IX there are 50 students 10 % students failed. How many students passed to class X?

7. Victor gets 92 % marks in examinations. If these are 460 marks, find the maximum marks.

Solution:

Let the maximum marks be m

Then 92 % of m = 460

⇒ 92/100 × m = 460

⇒ m = (460 × 100)/92

⇒ m = 46000/92

⇒ m = 500

Therefore, maximum marks in the examinations are 500

8. A company sold 31,250 televisions in 2020. In 2021, they sold 29,876 televisions. Calculate the percentage decrease in the number of televisions sold.

Solution:

The difference is

31250-29876=1374.

Therefore, the percentage change is

137431250×100%=4.4%.

Thus, the company sold 4.4% less televisions in 2021 than 2020.

9. Kevin buys a house for £250,000. After refurbishing, the house is now worth 10% more. He decides to sell the house. How much does the house cost now?

Solution

We first need to work out 10% of £250,000 to determine how much the price has increased by.

$10\% of 250,000 = 250,000 \div 10 = £25,000$. Thus, the cost of the house has increased by £25,000.

Thus, the new price of the house is
$£250,000 + £25,000 = £275,000$.

10. The population of a town is 8000. It increases by 10% during the first year and by 20% during the second year. Estimate the total population after 2 years.

Solution:

8000+800=8800, after first year

After second year, 8800+8800*1/5=10560 (ans)

11. Rahim spends 30% of his salary on food articles, and one-third of the remaining on transport. If he saves Tk. 2500 per month, which is the half of the balance after spending on food items and transport, find the monthly salary.

Solution: do yourself

Chapter 3

Interest

1. Find the compound interest on Rs 4500 for 3 years at 6% per annum. Interest being payable half yearly.
   Solution:
   $A=P(1+R/n100)2n$
   $=4500(1+6/200)6$
   $=4500(1.03)6$
   $=5373$
   Compound interest = Rs (5373 – 4500) = Rs 873

2. In a population of a town increases at 5% for the first 2 years, increases at 2% for the next 3 years , and then decreases at 2% for the next 2 years , what is the next % increase in the population in 7 years ?
   Solution:

3. On September 14, 2012, Bejos decided to pay back Jenie on his loan of $150000 at 9% simple interest. If he paid to Jenie the principle plus $1283.42 of interest, on calculate the day when he borrow the money from Jenie.

Solution:

$$t = \frac{I}{pr} = \frac{1283.42}{15000 \times \frac{0.09}{365}} = 347\ days$$

4. Raymond borrowed $450 at 0% interest. If he pays back 0.5% of the total amount every 7 days, beginning exactly 7 days after the loan was disbursed, and has thus far paid back $18, with the most recent payment made today, how many days ago did he borrow the money?

   Solution:
   0.5% of the amount borrowed is $(0.5/100)*450=\$2.25$
   (which is paid once every 7 days)

   Since the first installment of the loan that Raymond borrowed is paid 7 days later, Raymond pays the installment for $18/2.25=8$ weeks.

   As he made the most recent
   payment today, it is beginning of the 9th week since the money was lent.

   Therefore, Raymond would have borrowed the money before 8*7 = 56 days

5. Jack has 15000 Tk to invest. If he invests two-thirds of it into a high-yield savings account with an annual interest rate of 8%, compounded quarterly, and the other third in a regular savings account at 6% simple interest, how much does Jack earn after one year?

Solution:

Here, p=15000$\times \frac{2}{3} = 10000$, t=1year, r=8%

$$A = P\left(1 + \frac{r}{n \times 100}\right)^{nt}$$

$$10000\left(1 + \frac{8}{4 \times 100}\right)^{41} = 10824.32$$

Interest $I_1 = A - P = 10824.32 - 10000 = 824.32$

And for second savings,

p=15000$\times \frac{1}{3} = 5000$, t=1year, r=6%

Interest $I_2 = Pnr = 300$

Jack earn after 1 y =

$$I_1 + I_2 = 1124$$

6. $110000 is invested for 3 years at an annual interested rate of 10% per year simple interest. Calculate the value of the investment after this time.